



# DOWLEX™ 2027G

## Polyethylene Resin

### Overview

- Linear Medium Density Polyethylene
- For high speed, thin gauge applications
- Improved stiffness
- Complies with U.S. FDA 21 CFR 177.1520 (c) 3.2a.
- Complies with Canadian HPFB No Objection (With Limitations)
- Consult the regulations for complete details.

### Additive

- Antiblock: No
- Slip: No
- Processing Aid: No

| Physical                           | Nominal Value (English)    | Nominal Value (SI)      | Test Method |
|------------------------------------|----------------------------|-------------------------|-------------|
| Density                            | 0.941 g/cm <sup>3</sup>    | 0.941 g/cm <sup>3</sup> | ASTM D792   |
| Base Density <sup>1</sup>          | 0.941 g/cm <sup>3</sup>    | 0.941 g/cm <sup>3</sup> | Dow Method  |
| Melt Index (190°C/2.16 kg)         | 4.0 g/10 min               | 4.0 g/10 min            | ASTM D1238  |
| Films                              | Nominal Value (English)    | Nominal Value (SI)      | Test Method |
| Film Thickness - Tested            | 1.0 mil                    | 25 µm                   |             |
| Film Toughness                     |                            |                         | ASTM D882   |
| MD : 1.0 mil (25 µm)               | 1370 ft·lb/in <sup>3</sup> | 113 J/cm <sup>3</sup>   |             |
| TD : 1.0 mil (25 µm)               | 1500 ft·lb/in <sup>3</sup> | 124 J/cm <sup>3</sup>   |             |
| Tensile Strength                   |                            |                         | ASTM D882   |
| MD : Yield, 1.0 mil (25 µm)        | 2350 psi                   | 16.2 MPa                |             |
| TD : Yield, 1.0 mil (25 µm)        | 1560 psi                   | 10.8 MPa                |             |
| MD : Break, 1.0 mil (25 µm)        | 5660 psi                   | 39.0 MPa                |             |
| TD : Break, 1.0 mil (25 µm)        | 4060 psi                   | 28.0 MPa                |             |
| Tensile Elongation                 |                            |                         | ASTM D882   |
| MD : Break, 1.0 mil (25 µm)        | 580 %                      | 580 %                   |             |
| TD : Break, 1.0 mil (25 µm)        | 740 %                      | 740 %                   |             |
| Dart Drop Impact (1.0 mil (25 µm)) | 43 g                       | 43 g                    | ASTM D1709A |
| Elmendorf Tear Strength            |                            |                         | ASTM D1922  |
| MD : 1.0 mil (25 µm)               | 34 g                       | 34 g                    |             |
| TD : 1.0 mil (25 µm)               | 210 g                      | 210 g                   |             |
| Thermal                            | Nominal Value (English)    | Nominal Value (SI)      | Test Method |
| Vicat Softening Temperature        | 253 °F                     | 123 °C                  | ASTM D1525  |
| Melting Temperature (DSC)          | 261 °F                     | 127 °C                  | Dow Method  |
| Optical                            | Nominal Value (English)    | Nominal Value (SI)      | Test Method |
| Gloss (45°, 1.00 mil (25.4 µm))    | 86                         | 86                      | ASTM D2457  |
| Haze (1.00 mil (25.4 µm))          | 4.10 %                     | 4.10 %                  | ASTM D1003  |
| Extrusion                          | Nominal Value (English)    | Nominal Value (SI)      |             |
| Melt Temperature                   | 529 °F                     | 276 °C                  |             |

### Extrusion Notes

Fabrication Conditions For Cast Film:

- Screw Size: 3.5 in. (89 mm); 32:1 L/D
- Melt Temperature: 529°F (276°C)
- Screw Size: 2.5 in. (63.5 mm); 24:1 L/D
- Melt Temperature: 515°F (268°C)
- Chill Roll Temperature: 70°F (21°C)
- Line Speed: 500 fpm (152 m/min)

## Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

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<sup>1</sup> Base density is estimated using the assumption that every 1000 ppm of antiblock in the finished product raises the density of the polymer by 0.0006 g/cm<sup>3</sup>. Base density is the estimated density of the polymer if it did not contain any antiblock.

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